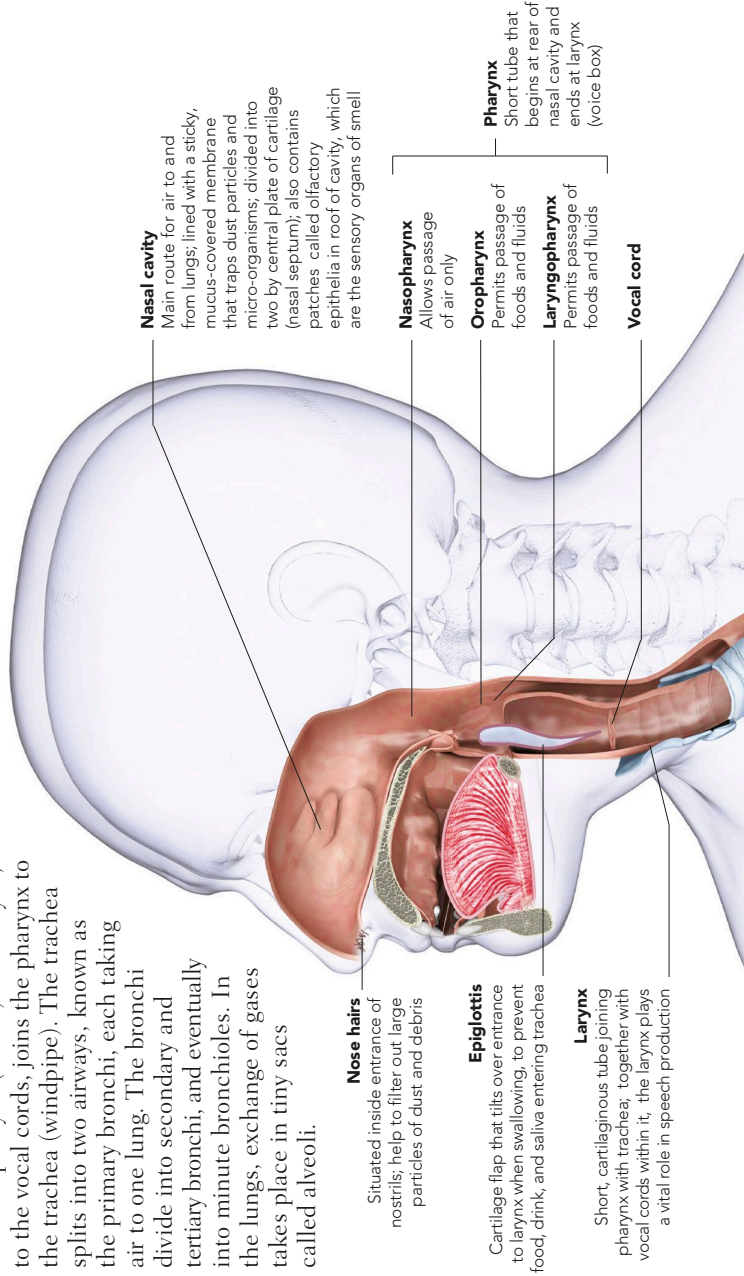


RESPIRATORY ANATOMY

THE RESPIRATORY SYSTEM, IN CLOSE CONJUNCTION WITH THE CIRCULATORY SYSTEM, IS RESPONSIBLE FOR SUPPLYING ALL BODY CELLS WITH ESSENTIAL OXYGEN AND REMOVING POTENTIALLY HARMFUL CARBON DIOXIDE FROM THE BODY.

Air enters the body mainly through the nostrils. The nostrils lead into the nasal cavity, which joins with the pharynx (throat). The larynx, home to the vocal cords, joins the pharynx to the trachea (windpipe). The trachea splits into two airways, known as the primary bronchi, each taking air to one lung. The bronchi divide into secondary and tertiary bronchi, and eventually into minute bronchioles. In the lungs, exchange of gases takes place in tiny sacs called alveoli.



Nasal cavity

Main route for air to and from lungs; lined with a sticky, mucus-covered membrane that traps dust particles and micro-organisms; divided into two by central plate of cartilage (nasal septum); also contains patches called olfactory epithelia in roof of cavity, which are the sensory organs of smell

Nasopharynx

Allows passage of air only

Oropharynx

Permits passage of foods and fluids

Laryngopharynx

Permits passage of foods and fluids

Pharynx

Short tube that begins at rear of nasal cavity and ends at larynx (voice box)

Vocal cord

Nose hairs

Situated inside entrance of nostrils; help to filter out large particles of dust and debris

Epiglottis

Cartilage flap that tilts over entrance to larynx when swallowing, to prevent food, drink, and saliva entering trachea

Larynx

Short, cartilaginous tube joining pharynx with trachea; together with vocal cords within it, the larynx plays a vital role in speech production